

provide optical sampling modules that support up to 80GHz bandwidth.

Fibre inspection and test tools.

Optical fibre inspection is critical for successful delivery of optical communication. Some specialised microscopes provide direct view of end-face cleanliness and core condition when inspecting fibre in the field. Vendors provide complete solutions for inspecting, cleaning and testing fibre connectors with a combination of all the necessary equipment that technicians need to ensure the integrity of optical connections.

Visual fault locator. It is an essential tool that quickly and easily locates problem areas in fibre cables. By pinpointing the exact location of fibre damage, technicians can diagnose, troubleshoot and fix the problem efficiently. These rugged units are a safe, economical and non-destructive way to identify live optical fibre. They use local detection technology, which employs a macro-blend method, eliminating the need to open the fibre at the splice point for identification.

Focus of T&M vendors

There have been several technological developments in optical communication systems. T&M technologies are being designed accordingly to address their needs.

"First, you must certify the physical media (for attenuation, PMD, CD, etc) to know whether the link would work with the desired technology (SDH, Ethernet). The second step is to test the specific technology and rate that you deployed. For example, in the case of SDH/SONET you can perform BER test, while for Ethernet/IP/FC you can run RFC2544 tests (up to layer 3) or some tests including higher layers,"

Some optical testing solution providers

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| 1. Agilent | 5. General Photonics |
| 2. Anritsu | 6. JDSU |
| 3. EXFO | 7. Rohde & Schwarz |
| 4. Fluke | 8. Tektronix |



Agilent's high-performance optical spectrum analyser

explains Damian Brazionis, product manager, Tecnous S.A.

According to Mombasawala, "Another important development is in the area of enhancing the optical communication capacity. Due to the bandwidth limitation of the fibre at 10 Gbps, transmission at 40G and 100G is not easy. There are non-linear effects in the fibre and components that start influencing the system. To counter this, a new transmission method called 'optical modulation' and similar to the RF modulation scheme is designed." Systems based on optical modulation require special test equipment like modulation analyser that allows various tests based on I-Q modulation for 40G and 100G systems.

To provide uninterrupted telecommunication link, the fibre-optic system requires continuous monitoring. Taws shares, "Test equipment can be integrated into an automated monitoring system and connected to a network operation centre. The system monitors the network continuously, alerting technicians and managers of faults as they occur."

"New remote fibre test systems are composed from a series of remote test units, a central server and multiple client stations, and machines using associated Web-based applications. Fibres are monitored in real time, 24x7 according to user-programmable schedules. Data polled from RTUs in the field is mapped to the central database and combined with routing records and geographical information, enabling maintenance teams to access

precise fault-location details," adds Taws.

As the technologies continuously evolve, there is a need for sustainable test systems. Tripathi shares, "Older equipment are upgraded to meet the new generation's network requirements. At some

point, however, really old equipment are beyond upgradability for today's applications. This is when the telecom companies have to opt for newer equipment."

Optical network in India

India is already making significant use of fibre-optics, with a high growth expected in the next few years. There are a number of telecom service providers in India with optical networks, several manufacturers, system integrators and service companies.

"India has approximately 1,000,000 km of optical fibre network operational using GPON, SDH and DWDM as a choice of technologies for transmission of information," informs Sandeep Gupta, vice president, Ericsson India.

"Many mobile operators own India-wide optical network. Optical fibre has reached the villages in India. This will help to meet the broadband demand of the market," shares Tripathi.

Mombasawala adds, "The digital microwave radio links used earlier by the operators are transitioning towards the optical-fibre links now. This way, there is deep penetration of optical networks within the country." All the overseas communication links too are now based on fibres.

As the demand for high-speed data services increases, there lies a huge requirement of reliable optical networks and reliable optical testing solutions. ●

The author is a senior technology journalist at EY